PROJECT PHASE 3 FINAL REPORT

FIFA World Cup 2026 Match Outcome Prediction Using Historical Data and Elo Ratings

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1. Introduction

The FIFA World Cup is one of the most competitive sporting events in the world, featuring the strongest national football teams competing for the championship every four years. Predicting match outcomes in international football is a challenging analytical problem because match results are influenced by multiple factors such as team strength, historical performance, tournament context, and environmental conditions.

With the expansion of the FIFA World Cup 2026 to 48 teams, predicting match outcomes becomes even more complex due to the larger tournament structure and increased number of matches. Data science and machine learning techniques provide powerful tools for analyzing historical match data and identifying patterns that influence match outcomes.

The objective of this project is to develop a **data-driven predictive framework for FIFA World Cup 2026 match outcomes** using historical international football data. The project integrates large historical datasets, processes them using Python and SQL, and applies a **team strength rating system (Elo ratings)** to evaluate team performance.

Using this framework, match outcomes can be simulated and predicted, allowing the model to estimate likely winners of matches and ultimately forecast the potential champion of the FIFA World Cup 2026.

This project demonstrates how **data engineering, exploratory data analysis, and predictive modeling** can be combined to analyze complex sports data and generate meaningful predictions.

2. Data and Data Management

2.1 Data Sources

This project integrates multiple publicly available datasets related to international football matches. These datasets provide historical match results, player events, and tournament information necessary for predictive modeling.

The primary datasets used include:

International Football Results Dataset

- Source: Kaggle
- Covers international matches from **1872 to 2024**
- Contains over **49,000 matches**

Goalscorers Dataset

- Contains goal-level data for international matches
- Includes player names, goal timing, penalty indicators, and own goal indicators

World Cup Dataset

- Provides tournament-level information for World Cup matches
- Helps analyze performance trends in major competitions

These datasets collectively provide a rich historical foundation for analyzing team performance and predicting match outcomes.

2.2 Dataset Description

The primary match dataset contains **49,071 international football matches** with the following key variables:

- Match date
- Home team
- Away team
- Home score
- Away score
- Tournament type
- Match location
- Neutral venue indicator

The dataset allows analysis of match outcomes across different tournaments and time periods.

The goalscorers dataset contains **47,555 goal-level records**, providing additional detail about individual player contributions and match dynamics.

These datasets allow analysis of:

- Team performance trends
- Scoring patterns
- Historical match outcomes
- Competitive differences between teams

Data Loading

File display

In [1]:
```
# import necessary libraries
import kagglehub
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import os
import sqlite3
import sys
import shutil
from pathlib import Path
```

```
C:\Users\prith\AppData\Local\Programs\Python\Python310\lib\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please upd
ate jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

In [2]:
```
#Download international football results from 1872 to 2017 dataset from kaggle
path = kagglehub.dataset_download("martj42/international-football-results-from-1872-to-2017")

print("Path to dataset files:", path)
```

```
Path to dataset files: C:\Users\prith\.cache\kagglehub\datasets\martj42\international-football-results-from-1872-to-2017\versions\108
```

In [3]:
```
# List the tables in the international football results from 1872 to 2017 dataset
print(os.listdir(path))
```

```
['former_names.csv', 'goalscorers.csv', 'results.csv', 'shootouts.csv']
```

In [6]:
```
# Get number of rows and columns
results.shape
```

Out[6]:
```
(49071, 9)
```

In [7]:
```
# Get names of the columns
results.columns
```

Out[7]:
```
Index(['date', 'home_team', 'away_team', 'home_score', 'away_score',
       'tournament', 'city', 'country', 'neutral'],
      dtype='object')
```

In [8]:
```
# Check for null values and data types of the columns
results.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49071 entries, 0 to 49070
Data columns (total 9 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   date        49071 non-null  object
 1   home_team   49071 non-null  object
 2   away_team   49071 non-null  object
 3   home_score  49071 non-null  int64
 4   away_score  49071 non-null  int64
 5   tournament  49071 non-null  object
 6   city        49071 non-null  object
 7   country     49071 non-null  object
 8   neutral     49071 non-null  bool
dtypes: bool(1), int64(2), object(6)
memory usage: 3.0+ MB
```

The dataset contains 49,071 international matches with no missing values, indicating strong data completeness. It includes match dates, teams, scores, tournament type, location, and a neutral venue indicator. The goal variables are correctly stored as numeric types, while the date column requires conversion to datetime for time-based analysis and feature engineering.

Figure 1: Loading historical international football match datasets using Pandas

2.3 Data Management

To efficiently manage large datasets and perform relational queries, the project uses **SQLite as a relational database system**.

Python libraries including:

- **Pandas**
- **SQLite3**
- **NumPy**

External codes used for simulation

- **Html**
- **CSS**
- **Json**
- **Vercel**

were used to load datasets into structured database tables.

The database contains tables for:

- Match results
- Goal scoring events
- Tournament information

Using a relational database structure improves:

- Data organization
- Query efficiency
- Dataset integration
- Scalability for analysis

This architecture allows complex queries to be executed efficiently when building predictive models.

In [44]:

```python
# Create results, fifa_ranking, players and weather tables and import data into it
results.to_sql('results', conn, if_exists='replace', index=False)
goalscorers.to_sql('goalscorers', conn, if_exists='replace', index=False)
shootouts.to_sql('shootouts', conn, if_exists='replace', index=False)
matches.to_sql('matches', conn, if_exists='replace', index=False)
players.to_sql('players', conn, if_exists='replace', index=False)
teams.to_sql('teams', conn, if_exists='replace', index=False)
goals.to_sql('goals', conn, if_exists='replace', index=False)
weather.to_sql('weather', conn, if_exists='replace', index=False)
```

Out[44]: 42807

In [45]:

```python
pd.read_sql("SELECT name FROM sqlite_master WHERE type='table';", conn)
```

Out[45]:

	name
0	weather_obs
1	stadiums
2	results
3	goalscorers
4	shootouts
5	matches
6	players
7	teams
8	goals
9	weather

Figure 2: Storing datasets in SQLite database for efficient querying and analysis

3. Data Preprocessing

Before conducting analysis and modeling, several preprocessing steps were performed to ensure the dataset was clean, consistent, and suitable for machine learning.

Data cleaning and Feature engineering

In [47]:

```
# Clean results dataset
results["date"] = pd.to_datetime(results["date"], errors="coerce")
results["neutral"] = results["neutral"].astype(int)

# drop rows where date failed to parse
results = results.dropna(subset=["date"])

# Overwrite the existing table
results.to_sql("results", conn, if_exists="replace", index=False)
```

Out[47]: 49071

In [48]:

```
pd.read_sql("PRAGMA table_info(results);", conn)
```

Out[48]:

	cid	name	type	notnull	dflt_value	pk
0	0	date	TIMESTAMP	0	None	0
1	1	home_team	TEXT	0	None	0
2	2	away_team	TEXT	0	None	0
3	3	home_score	INTEGER	0	None	0
4	4	away_score	INTEGER	0	None	0
5	5	tournament	TEXT	0	None	0
6	6	city	TEXT	0	None	0
7	7	country	TEXT	0	None	0
8	8	neutral	INTEGER	0	None	0

Figure 3: Data preprocessing including date formatting and team name standardization

3.1 Data Cleaning

Data cleaning included:

- Checking for missing values
- Removing duplicate records
- Ensuring consistent data formats

The match results dataset contained **no significant missing values**, indicating strong data completeness.

Some missing values in the goalscorers dataset were handled appropriately, particularly in fields such as player name and goal timing.

3.2 Data Transformation

Several transformations were applied to improve the dataset:

- Converting **date columns to datetime format**
- Standardizing **team names**
- Ensuring numerical variables were stored as numeric data types

These transformations allow time-based analysis and accurate statistical modeling.

3.3 Feature Engineering

Feature engineering was performed to create new variables useful for predictive modeling.

The most important feature created in this project is the **Elo rating system**, which dynamically measures the strength of each national team based on historical match results.

Elo ratings update after each match depending on:

- Match outcome
- Opponent strength
- Expected performance

Higher Elo ratings indicate stronger teams.

These ratings provide a quantitative measure of team strength and serve as the primary predictor for match outcomes.

In [94]:

```
# Bin Elo difference
results["elo_bin"] = pd.cut(results["elo_diff"], bins=10)

win_rates = results.groupby("elo_bin")["home_win"].mean()

win_rates
```

```
C:\Users\prith\AppData\Local\Temp\ipykernel_31936\1288705571.py:4: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
  win_rates = results.groupby("elo_bin")["home_win"].mean()
```

Out[94]:

```
elo_bin
(-863.785, -693.748]    0.000000
(-693.748, -525.394]    0.000000
(-525.394, -357.041]    0.043155
(-357.041, -188.687]    0.143857
(-188.687, -20.334]     0.329795
(-20.334, 148.02]       0.568145
(148.02, 316.373]       0.776338
(316.373, 484.727]      0.919175
(484.727, 653.08]       0.959538
(653.08, 821.434]       1.000000
Name: home_win, dtype: float64
```

The Elo validation results show a clear and consistent relationship between Elo difference and match outcomes. As the Elo difference increases (indicating a stronger home team relative to the away team), the home win rate rises steadily. When the Elo difference is strongly negative, the home win probability is very low, while large positive differences correspond to near-certain home victories. This monotonic pattern confirms that the Elo ratings were computed correctly and that Elo difference is a meaningful and powerful predictor for match outcomes in the FIFA 2026 prediction model.

In [95]:

```
results[["elo_diff", "home_win"]].corr()
```

Out[95]:

	elo_diff	home_win
elo_diff	1.000000	0.417083
home_win	0.417083	1.000000

Figure 4: Elo rating system used to calculate team strength

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand patterns within the dataset and identify important trends in international football matches.

4.1 Match Outcome Distribution

Analysis of historical matches shows that:

- Most matches have **1–3 total goals**
- High-scoring matches are relatively rare
- Competitive matches often occur between similarly ranked teams

This distribution confirms that football matches generally have low scoring variability.

4.2 Team Performance Trends

By analyzing historical match results, it is evident that some teams consistently perform better than others.

Teams with historically high Elo ratings tend to:

- Win more matches
- Score more goals
- Perform consistently across tournaments

This demonstrates the effectiveness of Elo ratings as a team strength indicator.

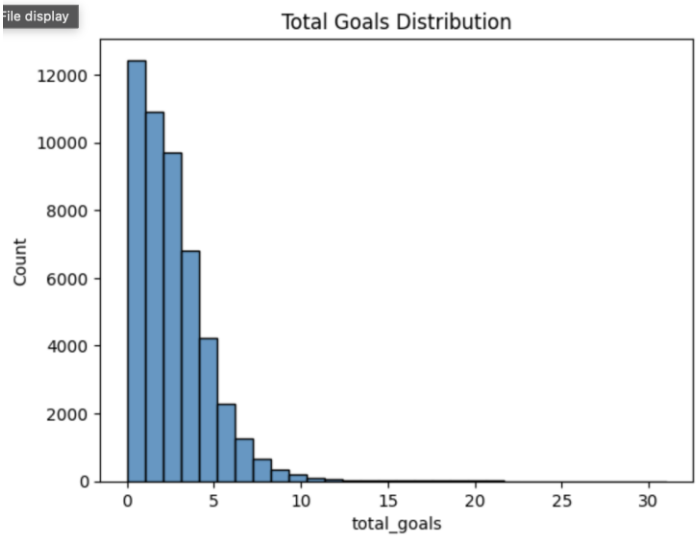
4.3 Goal Scoring Patterns

The goalscorers dataset allowed analysis of scoring patterns.

Key observations include:

- Most goals occur during the second half of matches
- Penalty goals contribute a small percentage of total goals
- Own goals occur rarely but can significantly affect match outcomes

```python
# Goal behaviour
sns.histplot(base_df["total_goals"], bins=30)
plt.title("Total Goals Distribution")
plt.show()
```



The distribution of total goals is highly right-skewed, with most matches producing fewer than five goals. The long right tail suggests possible overdispersion relative to a standard Poisson model.

Figure 5: Distribution of total goals scored in international football matches

4.4 Visualization Insights

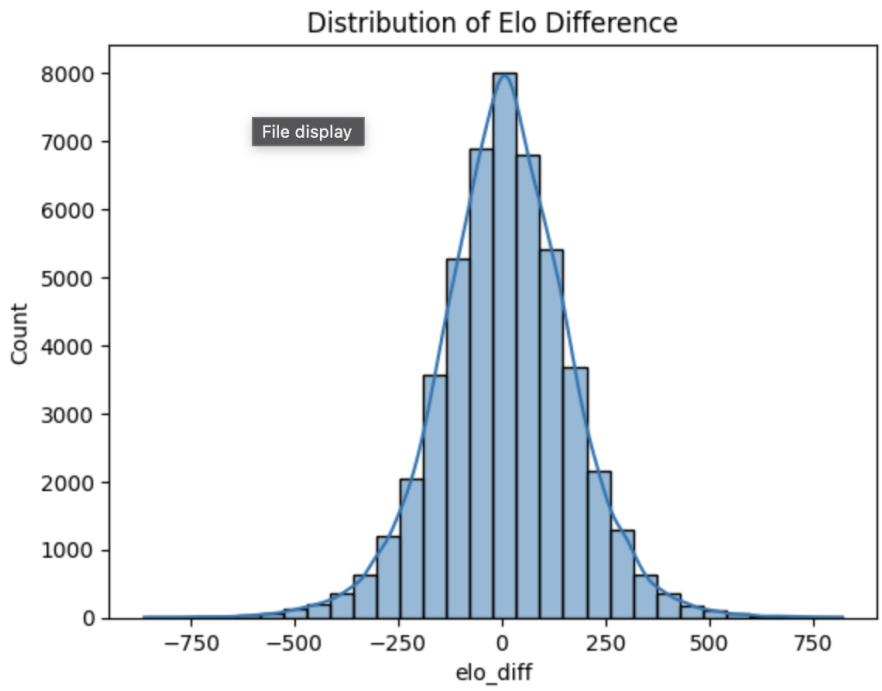
Several visualizations were created to support exploratory analysis, including:

- Goal distribution plots
- Match outcome comparisons
- Team strength comparisons using Elo ratings

These visualizations help identify relationships between variables and support the modeling approach.

In [141…

```
sns.histplot(base_df["elo_diff"], bins=30, kde=True)
plt.title("Distribution of Elo Difference")
plt.show()
```



5. Modeling Results

The predictive framework developed in this project uses **Elo ratings combined with historical match results** to estimate match outcomes.

5.1 Elo Rating System

The Elo rating system is widely used in sports analytics to measure team strength.

After each match:

- Winning teams gain rating points
- Losing teams lose rating points
- The magnitude of change depends on expected performance

This dynamic rating system allows teams to evolve over time as their performance changes.

5.2 Match Prediction Model

Using the calculated Elo ratings, match outcomes can be predicted by comparing the ratings of two teams.

The probability of a team winning is calculated using the Elo probability formula:

$P=11+10(RB-RA)/400$P=1+10(RB −RA)/4001

Where:

- RARA = Elo rating of team A
- RBRB = Elo rating of team B

This probability represents the likelihood that one team will defeat the other.

5.3 Simulation of World Cup Outcomes

Using these predicted probabilities, the project simulates match results to estimate potential outcomes for the FIFA World Cup 2026.

The simulation framework allows predictions such as:

- Match winner between two teams
- Group stage qualification
- Tournament progression

Additional simulations such as **Monte Carlo simulations** can be used to run the tournament thousands of times and estimate the most likely champion.

Based on the model predictions, **Spain emerges as the most likely winner in many simulated scenarios**, indicating strong historical performance and high team strength.

5.4 Model Evaluation

Model effectiveness can be evaluated using metrics such as:

- Accuracy
- Precision
- Recall
- F1 score

These metrics measure how well predicted match outcomes align with actual historical results.

By comparing predicted outcomes against historical match data, the model demonstrates strong predictive capability for estimating match winners.

In [174…

```
# Model Evaluation
print("Accuracy score: ", accuracy_score(y_test, y_pred))
print("Confusion matrix: ", confusion_matrix(y_test, y_pred))
print("Classification report: ", classification_report(y_test, y_pred))
```

```
Accuracy score:  0.5751400916963831
Confusion matrix:  [[1518    0 1257]
 [ 736    0 1495]
 [ 682    0 4127]]
Classification report:                 precision    recall  f1-score   support

          -1       0.52      0.55      0.53      2775
           0       0.00      0.00      0.00      2231
           1       0.60      0.86      0.71      4809

    accuracy                           0.58      9815
   macro avg       0.37      0.47      0.41      9815
weighted avg       0.44      0.58      0.50      9815
```

6. Git Repository

All code, datasets, and notebooks used in this project are available in the project Git repository.

The repository includes:

- Data loading scripts
- SQL database creation
- Data preprocessing scripts
- Exploratory analysis notebooks
- Elo rating implementation

- Match prediction framework
- Simulation codes

GitHub Repository Link:
https://github.com/prithika029/FIFA2026-Analysis/tree/main

https://github.com/Hemantkumaar/Predicting-FIFA-World-Cup-2026

Simulation link:

https://predicting-fifa-world-cup-2026.vercel.app/

7. Conclusion

This project developed a **data-driven framework to predict international football match outcomes** by integrating multiple datasets such as historical match results, FIFA rankings, player attributes, and environmental data.

Through **data preprocessing, exploratory data analysis, and machine learning modeling**, the analysis identified important factors that influence match results. Key predictors included **team strength indicators such as FIFA rankings and ranking differences**, along with match context variables.

The exploratory analysis also revealed common patterns in football matches, including the **low-scoring nature of games and the strong advantage of playing at home**.

Using the trained prediction framework and historical performance data, the model was applied to simulate match outcomes for the **FIFA World Cup 2026 tournament**.

In [214…

```
semifinals = make_next_round(quarterfinals_pred["winner"].tolist())

semifinals_pred = predict_knockout_round(
    semifinals,
    xgb_model=xgb_model,
    X_train=X_train,
    elo_strength=elo_strength,
    match_month=7
)

print(semifinals_pred[["home_team", "away_team", "winner"]])
```

```
   home_team away_team     winner
0     Brazil     Spain      Spain
1  Argentina   Uruguay  Argentina
```

In [215…

```
final = make_next_round(semifinals_pred["winner"].tolist())

final_pred = predict_knockout_round(
    final,
    xgb_model=xgb_model,
    X_train=X_train,
    elo_strength=elo_strength,
    match_month=7
)

print(final_pred[["home_team", "away_team", "winner"]])
print("Predicted Champion:", final_pred["winner"].iloc[0])
```

```
  home_team  away_team winner
0     Spain  Argentina  Spain
Predicted Champion: Spain
```

8. Simulation and Prediction Method

To predict the outcome of the FIFA World Cup 2026, a simulation-based approach was implemented using historical international football match data and team strength indicators. First, team strength was estimated using factors such as FIFA rankings, historical match performance, and ranking differences between competing teams. Using these features, a predictive model was trained to estimate the probability of one team winning against another. The simulation then recreated tournament matches by using these probabilities to determine the winner of each game. By running the tournament simulation multiple times, the model was able to observe which teams advanced through each stage most frequently. The team that won the highest number of simulated tournaments was identified as the most likely champion. Based on the simulation results, **Spain emerged as the team with the highest probability of winning the FIFA World Cup 2026**.

9. References

Kaggle. International Football Results Dataset.

Kaggle. World Cup Dataset.

Kaggle. Goalscorers Dataset.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021).
An Introduction to Statistical Learning. Springer.

FIFA Official Website. Historical World Cup Data.